

Primitive Data Types in Java

Java provides **8 primitive data types** to represent simple values like numbers, characters, and logical values. Primitive types are not objects; they store the actual value directly in memory, making them more efficient for basic operations.

1. Overview of Primitive Data Types

Data Type	Size	Default Value	Description
byte	1 byte (8 bits)	0	Small integers (-128 to 127)
short	2 bytes (16 bits)	0	Medium-range integers (-32,768 to 32,767)
int	4 bytes (32 bits)	0	Default for integers (-2^{31} to $2^{31} - 1$)
long	8 bytes (64 bits)	0L	Large integers (-2^{63} to $2^{63} - 1$)
float	4 bytes (32 bits)	0.0f	Decimal numbers (single precision)
double	8 bytes (64 bits)	0.0d	Decimal numbers (double precision)
char	2 bytes (16 bits)	'\u0000'	Single Unicode character (e.g., 'A')
boolean	1 bit	false	Logical values (true or false)

2. Detailed Explanation of Each Data Type

1. byte

- Size:** 1 byte (8 bits)
- Range:** -128 to 127
- Usage:** Useful for memory-constrained systems or when working with streams of data from devices like sensors.

Example:

```
public class ByteExample {  
    public static void main(String[] args) {  
        byte smallNumber = 100;  
        System.out.println("Byte Value: " + smallNumber);  
    }  
}
```

2. short

- **Size:** 2 bytes (16 bits)
- **Range:** -32,768 to 32,767
- **Usage:** Used for saving memory in large arrays where the range of values is known.

Example:

```
public class ShortExample {  
    public static void main(String[] args) {  
        short year = 2024;  
        System.out.println("Short Value: " + year);  
    }  
}
```

3. int

- **Size:** 4 bytes (32 bits)
- **Range:** -2,147,483,648 to 2,147,483,647
- **Usage:** Default choice for integer values.

Example:

```
public class IntExample {  
    public static void main(String[] args) {  
        int population = 1000000;  
        System.out.println("Int Value: " + population);  
    }  
}
```

4. long

- **Size:** 8 bytes (64 bits)
- **Range:** -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
- **Usage:** Used when an integer's range exceeds `int`.

Example:

```
public class LongExample {
    public static void main(String[] args) {
        long distanceToMars = 225000000000L;
        System.out.println("Long Value: " + distanceToMars);
    }
}
```

5. float

- **Size:** 4 bytes (32 bits)
- **Range:** Approximately $\pm 3.40282347E+38F$
- **Usage:** Used for fractional values with single precision.

Example:

```
public class FloatExample {
    public static void main(String[] args) {
        float pi = 3.14f;
        System.out.println("Float Value: " + pi);
    }
}
```

6. double

- **Size:** 8 bytes (64 bits)
- **Range:** Approximately $\pm 1.79769313486231570E+308$
- **Usage:** Default for decimal numbers. Used for high precision calculations.

Example:

```
public class DoubleExample {
    public static void main(String[] args) {
        double euler = 2.718281828459045;
        System.out.println("Double Value: " + euler);
    }
}
```

```
}  
}
```

7. char

- **Size:** 2 bytes (16 bits)
- **Range:** Unicode values (0 to 65,535)
- **Usage:** Represents a single character or symbol.

Example:

```
public class CharExample {  
    public static void main(String[] args) {  
        char letter = 'A';  
        char unicodeChar = '\u0041'; // Unicode for 'A'  
        System.out.println("Char Value: " + letter);  
        System.out.println("Unicode Char: " + unicodeChar);  
    }  
}
```

8. boolean

- **Size:** 1 bit
- **Values:** `true` or `false`
- **Usage:** Represents logical values.

Example:

```
public class BooleanExample {  
    public static void main(String[] args) {  
        boolean isJavaFun = true;  
        System.out.println("Is Java Fun? " + isJavaFun);  
    }  
}
```

3. Primitive Data Type Memory Sizes

The size of each data type ensures predictable memory allocation and efficient performance.

4. Key Points to Remember

- Primitive types are not objects. For example, `int` is a primitive, while `Integer` is a wrapper class.
 - Default values are assigned only to instance and static variables. Local variables must be initialized before use.
 - The `boolean` type is used in conditional statements like `if` and loops.
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5. Comprehensive Example

```
public class PrimitiveTypes {
    public static void main(String[] args) {
        byte b = 10;
        short s = 100;
        int i = 1000;
        long l = 100000L;
        float f = 3.14f;
        double d = 2.71828;
        char c = 'J';
        boolean flag = true;

        System.out.println("Byte: " + b);
        System.out.println("Short: " + s);
        System.out.println("Int: " + i);
        System.out.println("Long: " + l);
        System.out.println("Float: " + f);
        System.out.println("Double: " + d);
        System.out.println("Char: " + c);
        System.out.println("Boolean: " + flag);
    }
}
```

Output:

```
Byte: 10
Short: 100
Int: 1000
Long: 100000
Float: 3.14
Double: 2.71828
```

Char: J

Boolean: true